

MODEL QUESTION PAPER 1
S3 INSTRUMENTATION ENGINEERING
SUBJECT: CONTROL ENGINEERING

Time: 3 Hours

Max. Marks: 75

I: Part – A: Answer any five questions. Each question carries 3 marks. (5 x 3 = 15 Marks)

1. Write any three differences between open loop and closed loop system.
2. State initial value and final value theorems.
3. Define the terms transfer function and order of a control system.
4. State Mason's gain formulae.
5. Draw any three test input signals.
6. Define the terms absolute stability, relative stability and marginal stability.
7. Write the Laplace transform of $\sin(\omega t)$ and $\cos(\omega t)$

Part - B: Answer all questions. Each full question carries 15 marks. (4 x 15 = 60 Marks)

- II) (a) Determine the Laplace transform of the function $\cos(at)$ (7)
(b) Find the inverse Laplace transform of the function

$$X(s) = s / (s+2)(s+4) \quad (8)$$

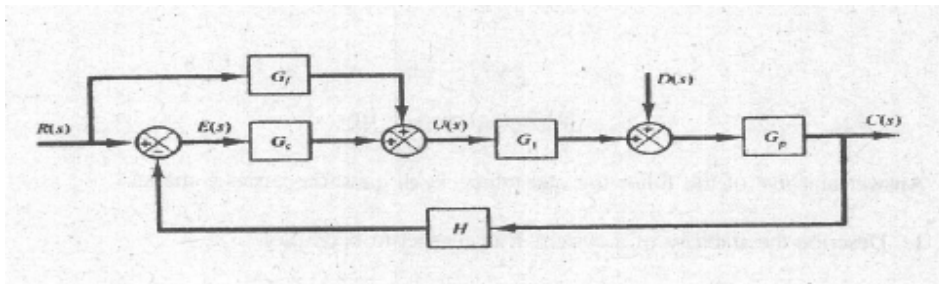
OR

- III) (a) Derive the Laplace transform of (i) e^{-at} and (ii) ramp input (6)
(b) Find the inverse Laplace Transform of the following functions (9)

$$(i) F(S) = \frac{1}{(S+1)(S+4)} \quad (ii) F(S) = \frac{1}{S^2 + 6S + 13}$$

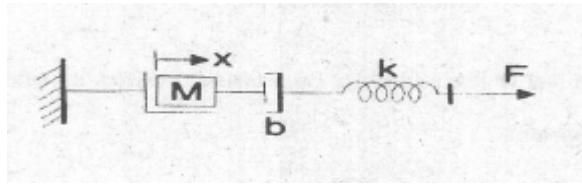
- IV) (a) Describe force current analogy. (6)

- (b) For the block diagram shown, determine the overall transfer function $C(s)/R(s)$ (9)

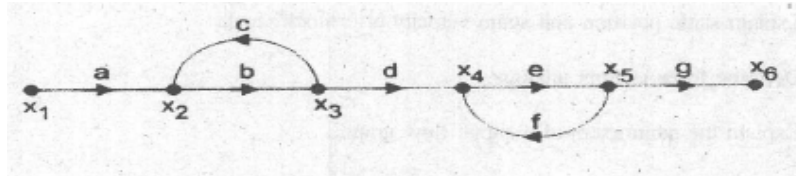


OR

- V) (a) Determine the mathematical model for the system shown (7)



- (b) Find the signal flow graph shown, determine the transfer function (8)



- VI) (a) Derive the response of a first order system to a unit ramp input. (8)

- (b) Explain static position and static velocity error coefficient. (7)

OR

- VII) (a) Find the response of first order system for unit step input. (8)

- (b) Derive the steady state error in terms of K_p , K_v and K_a for Type-0, Type-1 and Type-2 system when the input is unit ramp. (7)

- VIII) (a) Find the value of the controller gain K_c that makes the feedback control system of the following equation stable. (8)

$$10s^3 + 17s^2 + 8s + (1+K_c) = 0$$

- (b) With necessary steps draw the Bode plot for the function $1+sT$. (7)

OR

- IX) (a) Explain procedure to construct root locus. (8)

- (b) With necessary steps draw the Bode plot for the function $1/(1+sT)$. (7)

Scrutinized by:

Chairman:
Designation:
Institution:

Question Wise Anal

Course: **CONTROL ENGINEERING – QP 1**

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.02	U	3	9
I.2	M1.03	U	3	9
I.3	M2.01	U	3	9
I.4	M2.04	R	3	9
I.5	M3.01	R	3	9
I.6	M4.01	U	3	9
I.7	M1.03	U	3	9
II a.	M1.03	A	7	16
II b.	M1.03	A	8	18
III a.	M1.03	A	6	13.5
III b.	M1.03	A	9	20.5
IV a.	M2.04	U	6	13.5
IV b.	M2.04	A	9	20.5
V a.	M2.04	U	7	16
V b.	M2.04	A	8	18
VI a.	M3.01	U	8	18
VI b.	M3.02	U	7	16
VII a.	M3.01	U	8	18
VII b.	M3.03	U	7	16
VIII a.	M4.02	U	8	18
VIII b.	M4.03	A	7	16
IX a.	M4.04	U	8	18
IX b.	M4.04	A	7	16

Course: **CONTROL ENGINEERING – QP 1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 141 ($\pm 5\%$)	Type of Questions					
			Part A		Part B		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	35 $\pm$ 7	3	9	2	30	5	39
2	16	38 $\pm$ 7	2	6	2	30	4	36
3	14	33 $\pm$ 7	1	3	2	30	3	33
4	15	35 $\pm$ 7	1	3	2	30	3	33
Total	60	141	7	21	8	120	15	141

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	6	4.5
Understanding	74	52.5
Applying	61	43
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	141	100

MODEL QUESTION PAPER 2

S3 INSTRUMENTATION ENGINEERING

SUBJECT: CONTROL ENGINEERING

Time: 3 Hours

Max. Marks: 75

I: Part – A: Answer any five questions. Each question carries 3 marks. (5 x 3 = 15 Marks)

1. Define linear time invariant system with an example.
2. Derive the Laplace transform of e^{at} .
3. Define the terms Node, Branch and Loop of Signal flow graph
4. Describe any two block diagram reduction rules.
5. Write steady state errors for type 0, 1 and 2 systems for step, ramp and parabolic inputs.
6. Define the terms gain margin and phase margin.
7. What are asymptotes? Write the equation to find the angle of asymptotes.

Part - B: Answer all questions. Each full question carries 15 marks. (4 x 15 = 60 Marks)

- II. (a) Find the inverse Laplace transform of (8)

$$\frac{s^2 + 9s + 2}{(s-1)^2(s+3)}$$

- (b) Find the Laplace transform of $\sin at$ and $\cos at$ (7)

OR

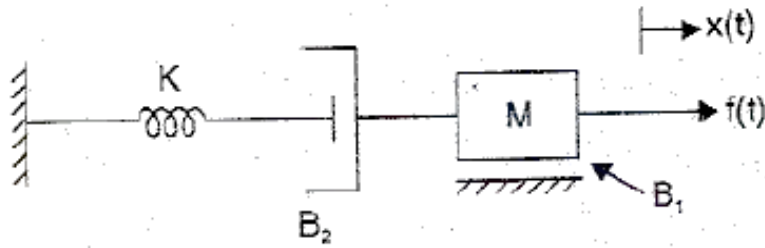
- III. (a) Find the inverse Laplace transform of (6)

$$\frac{1}{s(s-3)}$$

- (b) Use Laplace transform to solve the problem (9)

$$y'' + 3y' + 2y = e^{-t}, \quad y(0) = y'(0) = 0.$$

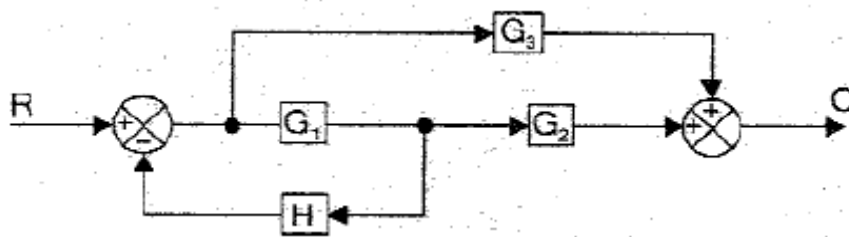
- IV. (a) Determine the transfer function of the system. (9)



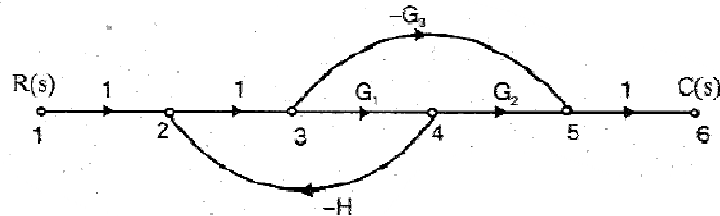
(b) Describe the Force- voltage analogy. (6)

OR

V. (a) Reduce the block diagram shown in the figure and find C/R. (8)



(b) The signal flow graph of a closed loop system is given. Determine the transfer function (7)



VI. Derive the steady state error for unit step, ramp and parabolic inputs for Types 0, 1 and 2 systems. (15)

OR

VII. (a) Derive the response of first order system for unit step input (9)

(b) For a unity feedback control system, the open loop transfer function (6)

$$G(s) = \frac{10(s+2)}{s^2(s+1)}, \text{ find static error constants.}$$

VIII. (a) Draw the Bode Plot for $G(s) = 1/(1+sT)$ (7)

(b) Determine the range of K for stability of unity feedback system (8)

whose open loop transfer function is $\frac{K}{s(s+1)(s+2)}$

OR

IX. (a) State the steps for constructing root locus. (8)

(b) Draw the Bode Plot for $G(s) = 1+sT$ (7)

Scrutinized by:

Surendran E

Question Wise Analysis

Chairman:

Designation:

Institution:

Course: **CONTROL ENGINEERING – QP 2**

Q. No	Module Outcome	Cognitive Level	Marks	Time (Min.)
I.1	M1.02	U	3	9
I.2	M1.03	U	3	9
I.3	M2.04	R	3	9
I.4	M2.04	U	3	9

I.5	M3.03	R	3	9
I.6	M4.03	U	3	9
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IV a.	M2.02	A	9	13.5
IV b.	M2.03	U	6	20.5
V a.	M2.04	A	8	16
V b.	M2.04	A	7	18
VI .	M3.03	U	15	34
VII a.	M3.01	U	9	18
VII b.	M3.02	A	6	16
VIII a.	M4.03	A	7	18
VIII b.	M4.03	A	8	16
IX a.	M4.04	U	8	18
IX b.	M4.04	A	7	16

Course : **CONTROL ENGINEERING – QP 2**

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Mark Distribution

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Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	6	4.5
Understanding	50	52.5
Applying	85	43
Analysing	0	0

Evaluating	0	0
Creating	0	0
Total	141	100